

TrES-3b

R-2801

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_130728 · created 2026-09-04T23:57:06+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2456501.7813 +/- 0.001 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.176 +/- 0.029
Transit depth (Rp/Rs)^2	3.11 +/- 1.04 [%]
Orbital Inclination (inc)	82.0 +/- 1.6
Airmass coefficient 1 (a1)	1.004 +/- 0.016
Airmass coefficient 2 (a2)	-0.004 +/- 0.023
Scatter in the residuals of the lightcurve fit is	1.59 %
Best Comparison Star	#1 - [301, 125]
Optimal Aperture	3.29
Optimal Annulus	10.4
Transit Duration (day)	0.053 +/- 0.018

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.176 $\pm$ 0.029 vs 0.16309 (deviation 0.45 σ)
Duration fit vs published	0.053 d vs 0.0567 d (deviation 0.2 σ)
Mid-time O–C	2.0 min
Depth SNR	6.1
Residual scatter	1.59 %

Light curve

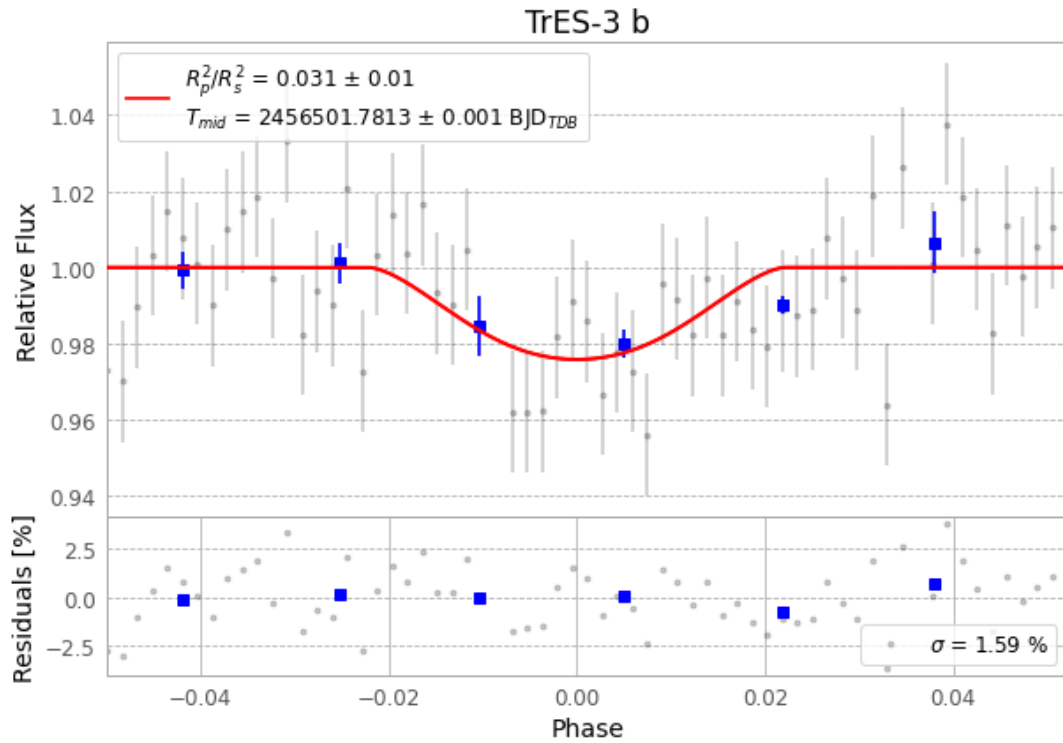


Figure 1 — FinalLightCurve_TrES-3 b_2013-07-27.png (SHA-256 464cbf8f5b59f673...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [373, 282], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 7df7f5136373f2e1...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

65 FITS frames (43 MB), TRES-3130728050612.FITS → TRES-3130728081812.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-130728.log (6978ceda3b17...), FinalLightCurve_TrES-3 b_2013-07-27.png (464cbf8f5b59...), FinalLightCurve_TrES-3 b_2013-07-27.csv (71012f59907b...)

Compute resources

Wall time 259.0 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-09-04 23:57Z.